

Not for submission. Read each reminder, then try the short practice that follows. Use the spaces for your calculations and explanations; the separate solutions let you check your work after each section.

1. Random variables, realizations, and averages

A *random variable* assigns a number to each possible outcome of a random process. We write X for the random variable and x for a possible *realization*, the number obtained in a particular outcome. If X is a die roll, observing a 4 gives the realization $x = 4$.

The *distribution* specifies the probabilities. For discrete X , write $p_X(x) = \Pr[X = x]$; these probabilities are nonnegative and sum to one.

A deterministic function g produces another random variable $g(X)$, with realization $g(x)$. The *expectation* averages possible outputs using the probabilities of their inputs:

$$\mathbb{E}[X] = \sum_x x p_X(x), \quad \boxed{\mathbb{E}[g(X)] = \sum_x g(x) p_X(x).}$$

Here x ranges over the possible values. An expectation is a fixed number determined by the distribution; it need not be a possible realization.

For example, if X is equally likely to be 0 or 2, then

$$\mathbb{E}[X] = 0 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 1, \quad \mathbb{E}[X^2] = 0^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{2} = 2.$$

Averaging squares differs from squaring the average: $\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2$ in this example.

An *indicator* $\mathbf{1}\{A\}$ is 1 on event A and 0 otherwise, so $\mathbb{E}[\mathbf{1}\{A\}] = \Pr[A]$. Linearity gives

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

for constants a, b, c , without requiring independence. All expectations in this handout are finite.

Guided practice

Let $\Pr[X = 0] = 1/2$, $\Pr[X = 1] = 1/3$, and $\Pr[X = 3] = 1/6$, and let $g(x) = x^2$.

- a. Explain the difference between X and x , and between $g(X)$ and $\mathbb{E}[g(X)]$. If the realization is $x = 3$, what is the realization of $g(X)$?

1. Random variables, realizations, and averages: practice (continued)

Let $\Pr[X = 0] = 1/2$, $\Pr[X = 1] = 1/3$, and $\Pr[X = 3] = 1/6$, and let $g(x) = x^2$.

- b. Write the weighted sums for $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$, then evaluate them. Does $\mathbb{E}[X^2]$ equal $(\mathbb{E}[X])^2$?

2. CDFs, densities, and expected payments

The *cumulative distribution function*, or CDF, is $F_X(t) = \Pr[X \leq t]$. Writing $X \sim F$ means that X has CDF F , without specifying its realization. The argument t is a fixed cutoff. A CDF is defined for every real t , including cutoffs outside the support, the range of possible values including its boundary.

If X has a *density* f_X , probabilities are areas:

$$\Pr[a < X \leq b] = \int_a^b f_X(x) dx = F_X(b) - F_X(a).$$

The density is nonnegative and integrates to one; $f_X = F'_X$ wherever this derivative exists. A density value is not a point probability: $\Pr[X = x] = 0$ for every fixed x , even if $f_X(x) > 1$. Thus changing a single endpoint from $<$ to $\leq$ does not change a probability for a variable with a density.

The distribution $U[a, b]$, for $a < b$, is uniform on $[a, b]$: its density is $1/(b - a)$ on that interval and zero elsewhere. For $X \sim U[0, 1]$, the CDF is t on $[0, 1]$, zero below 0, and one above 1.

To average a function of a variable with a density, replace the weighted sum by an integral:

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

We average the output $g(x)$ using the density of the input X . We do not need to find the distribution of $g(X)$ first. For example, if $X \sim U[0, 1]$, then $\mathbb{E}[X^2] = \int_0^1 x^2 dx = 1/3$.

For a fixed price r , a buyer pays r if their value X is at least r , and pays zero otherwise. The price is fixed; the payment depends on the random value. Indicators let us write such a payment as a single expression.

2. Guided practice

For both parts, use

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^2, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

- a. Differentiate to find the density on $(0,1)$. Compute $\Pr[X > 1/2]$ using the CDF. What is $\Pr[X = 1/2]$?

- b. A seller posts a fixed price $r \in (0,1)$. Fill in the realized payment and use the indicator identity to find its expectation:

$$g_r(X) = \text{_____} \mathbf{1}\{X \geq r\}, \quad \mathbb{E}[g_r(X)] = \text{_____} \Pr[X \geq r] = \text{_____}.$$

Check your answer by evaluating $\int_r^1 r f_X(x) dx$.

3. Conditioning and the tower property

A *joint distribution* specifies probabilities of pairs of realizations. Each interior entry below is $\Pr[X = x, Y = y]$. Sum a row or column to find the corresponding *marginal* probability.

	$Y = 0$	$Y = 1$	Total
$X = 0$	3/8	1/8	
$X = 1$	1/4	1/4	
Total			1

Conditioning restricts to outcomes consistent with the information and renormalizes their probabilities. For events A, B with $\Pr[B] > 0$,

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

Given $X = 0$, use only that row and divide each entry by its row total.

A *conditional expectation* averages using this conditional distribution:

$$m(x) = \mathbb{E}[Y \mid X = x] = \sum_y y \Pr[Y = y \mid X = x].$$

At a fixed x , $m(x)$ is a number. Before observing X , the expression $\mathbb{E}[Y \mid X]$ means the random variable $m(X)$. It records which conditional average we will use, not the realized value of Y .

The *tower property* averages these conditional averages using the probabilities of the conditioning cases:

$$\boxed{\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] = \sum_x m(x) \Pr[X = x].}$$

Independence is not needed. Discrete variables X, Y are *independent* if $\Pr[X = x, Y = y] = \Pr[X = x] \Pr[Y = y]$ for every pair. Equivalently, learning one does not change the distribution of the other. Independence allows $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$; linearity alone does not.

Guided practice

- Complete the marginal totals in the table. Divide by the row totals to find $\Pr[Y = 1 \mid X = 0]$ and $\Pr[Y = 1 \mid X = 1]$.

3. Conditioning and the tower property: practice (continued)

Continue with the same joint distribution:

	$Y = 0$	$Y = 1$
$X = 0$	$3/8$	$1/8$
$X = 1$	$1/4$	$1/4$

- b.** Since Y is either 0 or 1, its mean is its probability of being 1. Find $m(0)$ and $m(1)$, then compute $\mathbb{E}[m(X)]$. Check against $\mathbb{E}[Y]$ from the column totals.

- c.** Are X and Y independent? Compare $\Pr[X = 1, Y = 1]$ with $\Pr[X = 1] \Pr[Y = 1]$.

4. Fix a value, then average; reverse the bounds

Let X, Y be independent $U[0, 1]$ draws. If we learn $X = x$, then Y is still uniform. To average $X + Y$ given this information, replace X by the known number x : $\mathbb{E}[X + Y \mid X = x] = x + 1/2$. Then average over X : $\mathbb{E}[X + Y] = \int_0^1 (x + 1/2) dx = 1$.

These are the two steps for any function of the draws: hold x fixed and average over Y , then average the resulting function of x over X . For continuous draws, conditioning uses conditional densities; we do not divide by the zero point probability $\Pr[X = x]$. Independence means we can use the original density of Y .

Example: a random upper limit. Consider $\int_0^X 1 dt$. When $X = x$, this integral equals x . Before observing X , it is random. Since X is uniform, its average is

$$\mathbb{E} \left[\int_0^X 1 dt \right] = \int_{x=0}^1 \int_{t=0}^x 1 dt dx = \int_0^1 x dx = \frac{1}{2}.$$

The bounds describe the region $0 \leq t \leq x \leq 1$. If we fix t first, the allowed x values run from t to 1. Integrating over the same region in this order gives

$$\int_{t=0}^1 \int_{x=t}^1 1 dx dt = \int_0^1 (1 - t) dt = \frac{1}{2}.$$

The same step with other weights. If X instead has density f_X on $[0, 1]$ and we integrate a nonnegative function $h(t)$ up to X , the bounds stay the same:

$$\mathbb{E} \left[\int_0^X h(t) dt \right] = \int_0^1 h(t) \left(\int_t^1 f_X(x) dx \right) dt = \int_0^1 h(t)(1 - F_X(t)) dt.$$

The inner density integral is the probability that X reaches t . Nonnegativity justifies reversing the order.

Guided practice

Let X, Y be independent $U[0, 1]$ draws, and let $W = Y \mathbf{1}\{Y < X\}$.

- a. At a fixed $X = x$, the quantity W is Y when $Y < x$, and zero otherwise. Fill in the upper bound, evaluate, and average over X :

$$m(x) = \mathbb{E}[W \mid X = x] = \int_0^{\text{---}} y dy = \text{---}, \quad \mathbb{E}[W] = \int_0^1 m(x) dx = \text{---}.$$

4. Fix a value, then average; reverse the bounds: practice (continued)

As in part (a), X, Y are independent $U[0, 1]$ draws and $W = Y \mathbf{1}\{Y < X\}$.

- b. The calculation in part (a) integrates over $0 \leq y \leq x \leq 1$. If y is fixed first, which x values remain? Fill in the reversed bounds and evaluate to check the same answer:

$$\mathbb{E}[W] = \int_{y=0}^{\text{---}} \int_{x=\text{---}}^{\text{---}} y \, dx \, dy.$$

5. Tails, surplus, and independent events

For a nonnegative realization x , its height equals the length of the thresholds it exceeds: $x = \int_0^\infty \mathbf{1}\{x > t\} \, dt$. Averaging gives the *tail formula*:

$$\mathbb{E}[X] = \int_0^\infty \Pr[X > t] \, dt = \int_0^\infty (1 - F_X(t)) \, dt \quad \text{for } X \geq 0.$$

It adds up the probabilities of clearing each threshold. For $X \sim U[0, 1]$, this gives $\mathbb{E}[X] = \int_0^1 (1 - t) \, dt = 1/2$.

For a fixed $T \geq 0$, the *positive part* $(X - T)^+ = \max\{X - T, 0\}$ is the surplus above T . Shifting the thresholds gives

$$\mathbb{E}[(X - T)^+] = \int_T^\infty \Pr[X > t] \, dt.$$

For a buyer with value X facing price T , this is their expected utility when they can buy whenever $X \geq T$.

If an event A is independent of a random variable Z , then

$$\mathbb{E}[\mathbf{1}\{A\}Z] = \Pr[A]\mathbb{E}[Z].$$

An event determined only by one draw is independent of a function of another independent draw. This lets us separate the probability of reaching a buyer from their expected surplus. An event involving that buyer's own value need not be independent of their surplus.

Without independence, the useful identity is instead

$$\mathbb{E}[\mathbf{1}\{A\}Z] = \Pr[A]\mathbb{E}[Z \mid A] \quad (\Pr[A] > 0).$$

The unconditional average on the left includes a zero contribution outside A ; the conditional average on the right averages only within A .

5. Guided practice

Let X_1, X_2 be independent $U[0, 1]$ draws and fix $T \in (0, 1)$.

- a. Use the surplus formula to fill in and evaluate

$$\mathbb{E}[(X_2 - T)^+] = \int_T^1 \text{---} dt.$$

For $A = \{X_1 < T\}$, find $\Pr[A]$ and then $\mathbb{E}[\mathbf{1}\{A\}(X_2 - T)^+]$. State which two objects are independent.

- b. Instead let $C = \{X_2 \geq T\}$. Why is $\mathbf{1}\{C\}(X_2 - T)^+ = (X_2 - T)^+$ on every outcome? Find its expectation. Would multiplying $\Pr[C]$ by $\mathbb{E}[(X_2 - T)^+]$ give the same answer?

6. Order statistics: the largest and second-largest draws

An *order statistic* is a draw selected by its rank. For n draws, write X_k^n for the k -th highest: X_1^n is the maximum and X_2^n is the second-highest. The superscript labels the number of draws; it is not an exponent.

The abbreviation *i.i.d.* means independent and identically distributed. A distribution is *atomless* if every fixed point has probability zero. Independent draws from such a distribution tie with probability zero. Each index is equally likely to occupy each rank, by symmetry.

To find the CDF of a maximum $M = \max_i X_i$, first describe its event: $M \leq t$ exactly when every draw is at most t . For i.i.d. draws with CDF F ,

$$\Pr[M \leq t] = F(t)^n.$$

For the second-highest draw to be at most t , at most one draw can exceed t . Separate the cases of zero and one draw above the cutoff. Probabilities of disjoint cases add; independence lets us multiply probabilities within a case.

We can also compare sample sizes by deleting a uniformly random index, independently of the values. This leaves $n - 1$ i.i.d. draws. If we delete the largest draw, the remaining maximum is the original second-highest; otherwise, the maximum stays the same. Condition on the realized values first, then average.

Guided practice

Let X_1, X_2, X_3 be independent $U[0, 1]$ draws, with maximum $M = X_1^3$ and second-highest draw $S = X_2^3$.

- a. Find $F_M(t)$ for $0 \leq t \leq 1$, then use the tail formula:

$$\mathbb{E}[M] = \int_0^1 (1 - F_M(t)) dt = \underline{\hspace{1cm}}.$$

6. Order statistics: practice (continued)

Let X_1, X_2, X_3 be independent $U[0, 1]$ draws, with maximum $M = X_1^3$ and second-highest draw $S = X_2^3$.

- b.** For $0 \leq t \leq 1$, fill in the number of choices for the one draw above t :

$$F_S(t) = t^3 + ______ (1 - t)t^2.$$

Differentiate to find $f_S(t)$, then evaluate $\mathbb{E}[S] = \int_0^1 t f_S(t) dt$.

- c.** Delete a uniformly random index, independently of the values, and call the remaining maximum R . Given distinct realized values with largest m and second-largest s , fill in

$$\mathbb{E}[R \mid X_1 = x_1, X_2 = x_2, X_3 = x_3] = ______ m + ______ s.$$

Hint: Even after fixing all three values, each index is deleted with probability $1/3$. The conditional expectation averages only over this choice.

Then average over the draws using the tower property: the averages of m and s are $\mathbb{E}[M]$ and $\mathbb{E}[S]$. Use parts (a)–(b) to find $\mathbb{E}[R]$.